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UniCare: An Interoperable Electronic Health Record (EHR) Platform

24MCAR295 - Software Project Part 1

SYSTEM STUDY

Scrum Master

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Requirement Gathering

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1. Project Overview:

Briefly describe your project. What problem does it aim to solve, and what are its main objectives?

Project Title:

UniCare: Interoperable Electronic Health Record (EHR) Platform

Project Description:

UniCare is a centralized Electronic Health Record (EHR) platform designed to securely maintain a patient's Unified Patient Health Record (UPHR) across participating healthcare providers using a unique Health ID. The platform addresses the problem of fragmented medical records by enabling authorized healthcare professionals to securely access a patient's essential medical history, including previous prescriptions, laboratory reports, diagnostic reports, allergies, diagnoses, and long-term medication history through a role-based, consent-driven access mechanism.

Hospitals that wish to use UniCare can register on the platform and, upon approval by the Super Administrator, become participating healthcare providers. Patients are registered by authorized receptionists and receive a unique Health ID, allowing their medical records to be securely linked across all participating hospitals.

The platform enables patients to securely log in through a web portal to view their medical history, generate a Digital Health Card, upload laboratory and diagnostic reports obtained from external diagnostic centers, view digital prescriptions, and book appointments with participating hospitals. Doctors can search patients using the Health ID, access authorized medical records, generate digital prescriptions, upload laboratory and diagnostic reports, update patient diagnoses, and control whether medical records are shared with other participating hospitals. Hospital



administrators can manage doctors, receptionists, departments, appointments, and hospital users, while the Super Administrator manages hospital registration requests, participating hospitals, and overall platform administration.

The proposed system is developed as a web-based platform using React.js, Django, and MySQL.

Problem Addressed

- Patient medical records are maintained independently by different hospitals, resulting in fragmented healthcare information.
- Patients are often required to carry physical prescriptions, laboratory reports, and diagnostic reports when visiting different healthcare providers.
- Doctors may not have access to a patient's previous diagnoses, allergies, laboratory reports, prescriptions, or long-term medication history during consultation.
- The absence of essential medical history can lead to repeated laboratory investigations, duplicate prescriptions, delayed treatment, and increased risk of medication errors.
- Healthcare providers lack a secure and standardized platform to access authorized essential patient medical records across participating hospitals while ensuring patient privacy and controlled information sharing.

Main Objectives

The primary objectives are:

- Create a centralized hospital management platform.
- Provide unique registration for hospitals.
- Generate a unique Patient ID for every patient.
- Enable secure sharing of patient medical history among participating hospitals.
- Eliminate duplicate patient records.
- Improve healthcare quality through faster access to medical records.
- Maintain high standards of data security and privacy.

System Scope:

Specify the extent to which the system is proposed. Is it for research purposes, a prototype, or a full-scale implementation?

The proposed system is developed as an academic prototype for research and demonstration purposes. It focuses on developing a unified Electronic Health Record (EHR) platform that securely maintains a patient's Unified Patient Health Record (UPHR) across participating healthcare providers using a unique Health ID.

The scope of the system includes hospital registration and approval, patient registration, unique Health ID generation, Digital Health Card generation, secure authentication, role-based access control, appointment management, digital prescriptions, laboratory and diagnostic report management, controlled medical record sharing, and web portals for patients, doctors, receptionists, hospital administrators, and super administrators. Patients can securely access their medical history through the patient web portal, while authorized doctors can search patients using the Health ID, access essential medical records, generate digital prescriptions, upload laboratory and diagnostic reports, and classify records as Shared or Local. Hospital administrators can manage hospital users, departments, doctors, receptionists, and appointments, while the Super Administrator manages hospital registration requests, participating hospitals, and overall platform administration.

The platform demonstrates secure sharing of essential patient medical information between participating hospitals while preserving patient privacy through role-based access control and consent-based record access.

Phase 2 extends the platform by introducing a dedicated Flutter-based patient mobile application with medicine reminders and healthcare notifications. Future enhancements also include the integration of Artificial Intelligence (AI) and Machine Learning (ML) modules, such as disease risk prediction, drug interaction detection, duplicate prescription detection, prescription simplification using Natural Language Processing (NLP), and healthcare analytics dashboards.



The proposed system serves as a proof-of-concept for an interoperable healthcare platform and provides a scalable foundation for future intelligent, secure, and patient-centric healthcare services.

2. Target Audience:

Identify the viewers or public who will be involved with the system. Are there specific user groups or stakeholders?

The proposed UniCare – Interoperable Electronic Health Record (EHR) Platform involves multiple stakeholders who interact with the system based on their roles and responsibilities.

1. Patients

Patients are the primary beneficiaries of the platform. They can securely access their Unified Patient Health Record (UPHR), view digital prescriptions, upload laboratory and diagnostic reports from external diagnostic centres, generate a Digital Health Card, and book appointments with participating healthcare providers.

2. Doctors

Doctors access authorized patient medical records using the patient's Health ID. They can review longitudinal medical history, record diagnoses, generate digital prescriptions, upload laboratory and diagnostic reports, and determine whether medical records should be shared with other participating hospitals.

3. Receptionists

Receptionists register patients, generate unique Health IDs, search existing patient records, and manage appointment scheduling for their respective hospitals.

4. Hospital Administrators

Hospital Administrators manage hospital operations within the UniCare platform, including doctor and receptionist accounts, departments, appointments, and hospital users. They are also responsible for maintaining hospital information.

5. Super Administrator

The Super Administrator oversees the entire UniCare platform by approving hospital registration requests, managing participating hospitals, monitoring platform activities, generating reports, and viewing platform analytics.

6. Participating Healthcare Providers

Hospitals and healthcare organizations that register and are approved to use UniCare become participating healthcare providers. They contribute to maintaining the Unified Patient Health Record and collaborate in providing continuous patient care through secure information sharing.

3. Modules:

List the key modules included in your system. What functionalities do they provide?

The proposed system consists of the following key modules:

Module 1: Authentication and User Management

Functionality:

- Secure login for all users
- Role-based access control
- Hospital registration request
- Hospital approval by Super Administrator
- User account management

Module 2: Hospital Management

Functionality:

- Register participating hospitals
- Manage hospital information
- Manage departments
- Manage doctors
- Manage receptionists

Module 3: Patient Management

Functionality:

- Patient registration
- Unique Health ID generation
- Digital Health Card generation
- Patient profile management
- Search patient using Health ID

Module 4: Appointment Management

Functionality:

- Book appointments
- Manage appointment schedules
- View appointment history
- Appointment status management

Module 5: Electronic Health Record (EHR) Management

Functionality:

- Maintain Unified Patient Health Record (UPHR)
- Record diagnoses
- Maintain patient visit history
- Store longitudinal medical history
- View authorized patient records

Module 6: Prescription Management

Functionality:

- Generate digital prescriptions
- Manage prescribed medicines
- View prescription history
- Share or restrict prescription visibility

Module 7: Laboratory and Diagnostic Report Management

Functionality:

- Upload laboratory reports
- Upload diagnostic scan reports
- View laboratory reports
- Control report sharing between participating hospitals

Module 8: Medical Record Sharing

Functionality:

- Share essential medical records
- Mark records as **Shared** or **Local**

- Consent-based record access
- Secure access by authorized healthcare providers

Module 9: Patient Portal

Functionality:

- View medical history
- View digital prescriptions
- Generate Digital Health Card
- Upload external laboratory reports
- Book appointments

Module 10: Administration and Analytics

Functionality:

- Manage participating hospitals
- Monitor platform activities
- View platform usage statistics
- Generate administrative reports
- Maintain audit logs

4. User Roles:

Identify the different user roles in your project. Who will interact with the system, and what permissions or access levels do they have?

The proposed UniCare platform consists of three primary user roles, each with specific responsibilities and access permissions.

1. Super Administrator

Responsibilities

- Approve hospital registration requests
- Manage participating hospitals
- Monitor platform activities
- View platform analytics
- Generate administrative reports
- Manage system-wide configurations

Access Level

2. Hospital Administrator

Responsibilities

- Manage hospital profile
- Register and manage doctor accounts
- Register and manage receptionist accounts
- Manage departments
- Manage appointment schedules
- Monitor hospital activities
- View hospital usage statistics

Access Level

- Full access to resources belonging to their own hospital only.

3. Receptionist

Responsibilities

- Register patients
- Generate unique Health IDs
- Search existing patients
- Book appointments
- Manage appointment schedules
- Update basic patient information

Access Level

- Access limited to patient registration and appointment management within their hospital.

4. Doctor

Responsibilities

- Search patients using Health ID
- Access authorized patient medical records
- View longitudinal patient history
- Record diagnoses
- Generate digital prescriptions
- Upload laboratory and diagnostic reports
- View patient-uploaded reports
- Mark medical records as **Shared** or **Local**

Access Level

- Access only to authorized patient records and medical information required for treatment.

5. Patient**Responsibilities**

- Login using Health ID
- View Unified Patient Health Record (UPHR)
- Generate Digital Health Card (E-Health Card)
- View digital prescriptions
- Book appointments

Access Level

- Access limited to their own healthcare records and personal information.

5. System Ownership:**Who owns the system? Is it an academic institution, an organization, or an individual?**

The proposed UniCare – Interoperable Electronic Health Record (EHR) Platform is developed as an academic research prototype as part of the Master of Computer Applications (MCA) curriculum. For the purpose of this project, the platform is assumed to be owned and managed by the UniCare Platform Administration, represented by the Super Administrator, who is responsible for approving hospital registration requests, managing participating hospitals, monitoring platform activities, and maintaining overall system administration.

Healthcare organizations that wish to use the platform must register and obtain approval from the Super Administrator before becoming participating healthcare providers. Once approved, each participating hospital is assigned a unique Hospital ID and can securely manage its healthcare operations through the UniCare platform while contributing to the Unified Patient Health Record

(UPHR). This ownership model ensures centralized platform administration while enabling secure collaboration among participating healthcare providers.

6. Industry/Domain:

Specify the industry or domain to which the system is related (e.g., healthcare, finance, transportation).

The proposed UniCare – Interoperable Electronic Health Record (EHR) Platform belongs to the Healthcare industry, with a focus on securely managing and maintaining a Unified Patient Health Record (UPHR) across participating healthcare providers. The platform supports digital healthcare services by enabling authorized access to patient medical information, improving continuity of care, reducing duplicate medical investigations, and enhancing coordination among participating hospitals.

Industry: Healthcare

Domain:

- Electronic Health Records (EHR)
- Health Information Systems (HIS)
- Digital Healthcare
- Healthcare Information Management
- Medical Record Management
- Interoperable Healthcare Systems

7. Data Collection Contacts:

Provide details of the person(s) you have contacted for data collection. Include their name, role, and contact information.

Name: Dr. Sneha John

Clinic Name: Heal Holistic Homeopathy

Place: Pampady,Kottayam

Contact Number: 8547744346

8. Questionnaire for Data Collection:

Create a questionnaire with at least 10 questions. Ensure it covers relevant aspects of your project. If applicable, attach additional documents (e.g., bill receipts, certificate models).

1. What difficulties do you face when patients visit your clinic without previous medical records?

Patients often fail to bring their previous prescriptions, laboratory reports, and diagnostic reports, making it difficult to understand their complete medical history and continue treatment effectively.

2. Do you think maintaining patient medical records in a centralized digital platform would improve continuity of patient care?

Yes. A centralized digital platform would allow doctors to access important patient information quickly and continue treatment more effectively.

3. If a patient discontinues treatment at one hospital and later visits another hospital or clinic, how helpful would it be if previous digital prescriptions and reports were immediately available?

It would be highly beneficial. Doctors could review the patient's previous treatment history without repeating unnecessary investigations or asking patients to recall their complete medical history.

4. Have you experienced situations where patients discontinue treatment and later visit another hospital or clinic?

Yes. This happens frequently. If their previous medical records are readily available, it would help doctors understand the patient's history and continue treatment without unnecessary repetition.

5. Which patient information should always be available during consultation?

Previous prescriptions, laboratory reports, diagnostic scan reports, allergy history, ongoing medications, and previous diagnoses should always be available.

6. Would a Unique Health ID help in identifying patients and retrieving their medical records?

Yes. A Unique Health ID would help doctors quickly identify patients and retrieve their medical records from the platform.

7. What additional feature would you recommend for such a healthcare platform?

A QR Code should be included in the Digital Health Card. By scanning the QR code, authorized healthcare professionals can quickly retrieve the patient's Health ID and access authorized medical records.

8. How important is patient consent before allowing another healthcare provider to access medical records?

Patient consent is essential. Medical records should only be accessible to authorized healthcare professionals with the patient's permission.

9. Would a patient mobile application be useful for managing healthcare information?

Yes. A mobile application would help patients easily access prescriptions, laboratory reports, medicine reminders, and appointment details whenever required.

10. Do you think an interoperable Electronic Health Record platform connecting participating healthcare providers would improve healthcare delivery?

Yes. Such a platform would improve continuity of care, reduce repeated investigations, and enable doctors to make better treatment decisions by accessing a patient's previous medical history.

Geotagged Photograph



FEASIBILITY ANALYSIS

Feasibility is defined as the practical extent to which a project can be performed successfully. To evaluate feasibility, a feasibility study is performed, which determines whether the solution considered to accomplish the requirements is practical and workable in the software. Information such as resource availability, cost estimation for software development, benefits of the software to the organization after it is developed and cost to be incurred on its maintenance are considered during the feasibility study. The results of the feasibility study should be a report that recommends whether or not it is worth carrying on with the requirements engineering and system development process.

If a system does not support the business objectives, it has no real value to the business. While this may seem obvious, many organisations develop systems which do not contribute to their objectives either because they don't have a clear statement of these objectives, because they fail to define the business requirements for the system or because other political or organisation factors influence the system procurement.

The objective of the feasibility study is to establish the reasons for developing the software that is acceptable to users, adaptable to change and conformable to established standards. Various other objectives of feasibility study are listed below.

- To analyse whether the software will meet organizational requirements
- To determine whether the software can be implemented using the current technology and within the specified budget and schedule
- To determine whether the software can be integrated with other existing software.

The information assessment phase identifies the information that is required to answer the three questions set out above. Once the information has been identified, you should question information sources to discover the answers to these questions. Some examples of possible questions that might ask:

1. How would the organisation cope if this system was not implemented?

2. What are the problems with current processes and how would a new system help alleviate these problems?
3. What direct contribution will the system make to the business objectives and requirements?
4. Can information be transferred to and from other organisational systems?
5. Does the system require technology that has not previously been used in the organisation?
6. What must be supported by the system and what need not be supported?

Types of Feasibility

Various types of feasibility that are commonly considered include technical feasibility, operational feasibility, and economic feasibility.

Technical Feasibility

Technical feasibility evaluates whether the proposed system can be developed using the available hardware, software, technologies, and technical expertise.

The proposed UniCare: Interoperable Electronic Health Record (EHR) Platform is technically feasible because it is developed using well-established and widely adopted technologies such as React.js for the web application, Django for backend development, and MySQL for database management. Flutter is planned for the patient mobile application in Phase 2. These technologies are open-source, stable, scalable, and supported by extensive documentation and active developer communities.

The proposed system can be developed using the available computing resources, software development tools, and technical expertise. The development team possesses the necessary knowledge to design and implement the platform. Since the project is developed as an academic prototype, multiple participating hospitals can be simulated within the UniCare platform to

demonstrate secure medical record sharing without requiring access to real hospital databases. Therefore, the proposed system is technically feasible.

Operational Feasibility

Operational feasibility determines whether the proposed system will effectively satisfy user requirements and improve existing healthcare processes.

The proposed UniCare platform addresses the problem of fragmented patient medical records by providing a unified Electronic Health Record (EHR) platform where participating hospitals can securely maintain and access authorized patient medical records using a Unique Health ID and patient consent.

Hospitals can register on the platform and, upon approval by the Super Administrator, become participating healthcare providers. Patients are registered by authorized receptionists and can securely access their medical records, Digital Health Card, digital prescriptions, and appointment information through the patient web portal. Doctors can retrieve authorized patient records, generate digital prescriptions, upload laboratory and diagnostic reports, record diagnoses, and classify medical records as Shared or Local. Hospital administrators can manage hospital users, departments, appointments, and hospital operations through the platform.

During requirement gathering, healthcare professionals indicated that such a platform would improve continuity of patient care, particularly when patients continue treatment at another participating hospital. They also recommended incorporating a QR Code into the Digital Health Card to facilitate faster patient identification and retrieval of authorized medical records. Therefore, the proposed system is operationally feasible.

Economic Feasibility

Economic feasibility evaluates whether the expected benefits of the proposed system justify the development cost.

The proposed UniCare platform is developed using open-source technologies including React.js, Django, MySQL, Visual Studio Code, Git, and GitHub, thereby minimizing software licensing costs. Since the system is developed as an academic prototype, the primary investment involves development time, testing, and available hardware resources. The Flutter framework required for the patient mobile application in Phase 2 is also open source, further reducing development costs.

The proposed platform has the potential to reduce duplicate medical investigations, minimize medication errors, improve continuity of patient care, reduce dependence on physical medical records, and enhance coordination among participating healthcare providers. By maintaining a Unified Patient Health Record (UPHR) and enabling secure access to authorized medical information, the platform improves healthcare efficiency and decision-making. These long-term benefits outweigh the development cost, making the proposed system economically feasible.